

- 5 When the mass is fully submerged in the water, the sum of the upthrust U acting on the mass and the tension by the string on the mass (which is T) equals to the weight of the mass, i.e. $M = U + T$

The reading on the weighing scale will be the sum of the weight of the beaker of water and the reaction force of the upthrust by the mass on the water. Hence,

$$\begin{aligned}\text{reading} &= W + U \\ &= W + (M - T)\end{aligned}$$

Answer: D

- 6 When the car is traveling at maximum speed, there is no net force acting on the car. Hence